

PROFESSIONAL LABORATORY REPORT

MITRE ATT&CK

Attack Stage Identification & Detection Mapping

A 20-Technique Case Study Across the Adversary Attack Lifecycle

Detection Engineering & Threat Intelligence Research

Prepared By

Khizar UI Islam

BS Software Engineering — SZABIST Islamabad

CEH Certified | Aspiring SOC Analyst L1

mrkhizar97@gmail.com | linkedin.com/in/khizarulislam79 | github.com/Khizar-malik97

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1. Executive Summary

This report documents a detection-engineering research exercise built on the MITRE ATT&CK® Framework. Rather than focusing on a single malware family, this lab took a technique-first approach: researching individual adversary techniques directly from the ATT&CK knowledge base, extracting their tactic classification and platform scope, and — critically — identifying the detection method and log sources a SOC analyst would use to catch each technique in a live environment.

Two techniques were studied in depth as worked examples — PowerShell abuse (T1059.001) and DLL Sideload (T1574.001) — before the same methodology was scaled across 20 distinct attack scenarios spanning the full adversary lifecycle, from Initial Access and Execution through Persistence, Lateral Movement, Credential Access, Discovery, and Impact. The result is a consolidated detection-mapping reference suitable for use as a SOC playbook input or detection-engineering baseline.

2. Lab Objective

The objective of this lab was to move beyond simply reading ATT&CK technique descriptions and instead build a practical, analyst-ready detection reference. This required identifying, for each technique, not only what the adversary does but how a defender would realistically detect it — which log source, which event ID, and which behavioral signal would surface the activity.

- Research individual ATT&CK techniques directly from the official knowledge base, capturing tactic, technique ID, and platform.
- For each technique, define a concrete detection method grounded in real log sources (Sysmon, Windows Event Logs, EDR/XDR telemetry, etc.).
- Complete two full worked case studies (PowerShell abuse and DLL Sideload) to validate the research-to-detection workflow.
- Scale the methodology to a 20-technique master table spanning the full attack lifecycle.
- Identify patterns across techniques that inform broader SOC detection strategy.

3. Case Study 1 — PowerShell Downloading from External Source (T1059.001)

The first case study examined how adversaries abuse PowerShell to execute malicious commands, download second-stage payloads, and automate post-exploitation activity. Research began on the official MITRE ATT&CK technique page for PowerShell.

Home > Techniques > Enterprise > Command and Scripting Interpreter > PowerShell

Command and Scripting Interpreter: PowerShell

Other sub-techniques of Command and Scripting Interpreter (13) ▾

Adversaries may abuse PowerShell commands and scripts for execution. PowerShell is a powerful interactive command-line interface and scripting environment included in the Windows operating system.^[1] Adversaries can use PowerShell to perform a number of actions, including discovery of information and execution of code. Examples include the `Start-Process` cmdlet which can be used to run an executable and the `Invoke-Command` cmdlet which runs a command locally or on a remote computer (though administrator permissions are required to use PowerShell to connect to remote systems).

PowerShell may also be used to download and run executables from the Internet, which can be executed from disk or in memory without touching disk.

A number of PowerShell-based offensive testing tools are available, including [Empire](#), [PowerSploit](#), [PoshC2](#), and [PSAttack](#).^[2]

PowerShell commands/scripts can also be executed without directly invoking the `powershell.exe` binary through interfaces to PowerShell's underlying `System.Management.Automation` assembly DLL exposed through the .NET framework and Windows Common Language Interface (CLI).^{[3][4][5]}

Procedure Examples

ID: T1059.001

Sub-technique of: T1059

① Tactic: [Execution](#)

① Platforms: [Windows](#)

Contributors: Mayuresh Dani, Qualys; Praetorian; Ross Brittain

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[Version](#) [Permalink](#)

Figure 1 — MITRE ATT&CK technique page for PowerShell (T1059.001), detailing tactic, platform, and adversary usage.

Attack Scenario	PowerShell Downloading from External Source
ATT&CK Tactic	Execution
Technique ID	T1059.001
Technique Name	Command and Scripting Interpreter: PowerShell
Platform	Windows

Explanation

This technique page identifies the technique ID, tactic (Execution), and supported platform (Windows), and explains how attackers abuse PowerShell to execute malicious commands, download payloads, and automate post-exploitation activities. Because PowerShell is a legitimate, pre-installed administrative tool, its abuse is difficult to distinguish from normal system administration without behavioral context.

Detection Strategy

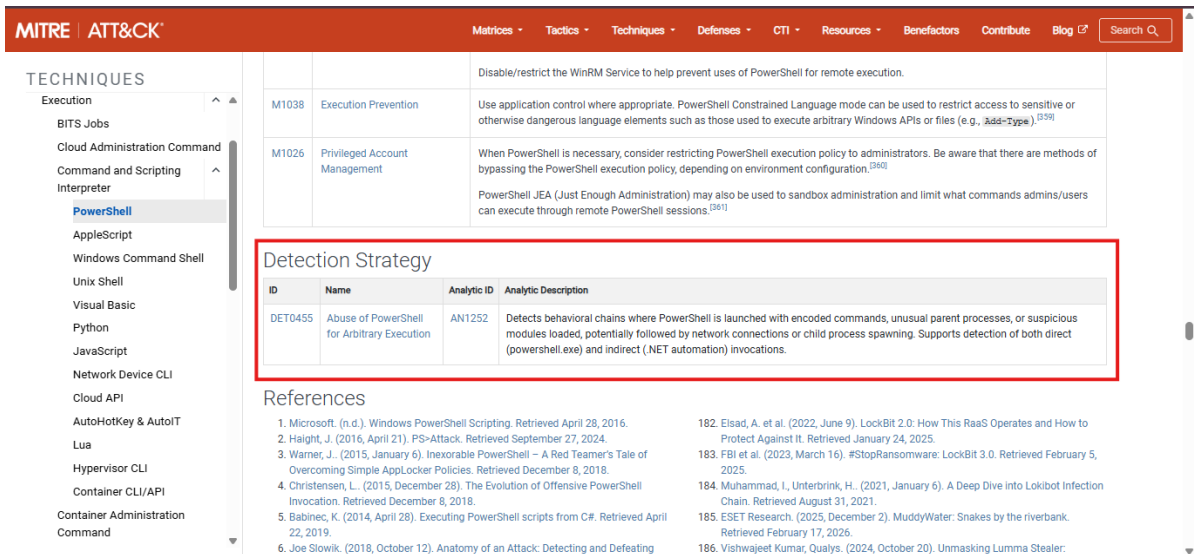


Figure 2 — ATT&CK detection guidance for PowerShell-based execution.

The detection method for this scenario centers on monitoring for suspicious PowerShell commands and script execution — particularly encoded (Base64) command-line arguments, unusual parent/child process relationships (e.g., Office applications spawning PowerShell), and script block logging.

Attack Scenario	ATT&CK Tactic	Technique ID	Detection Method	Relevant Logs
PowerShell Downloading from External	Execution	T1059.001	Monitor suspicious PowerShell commands and script execution	PowerShell Operational Logs, Sysmon Event ID 1, Windows Security Event ID 4688

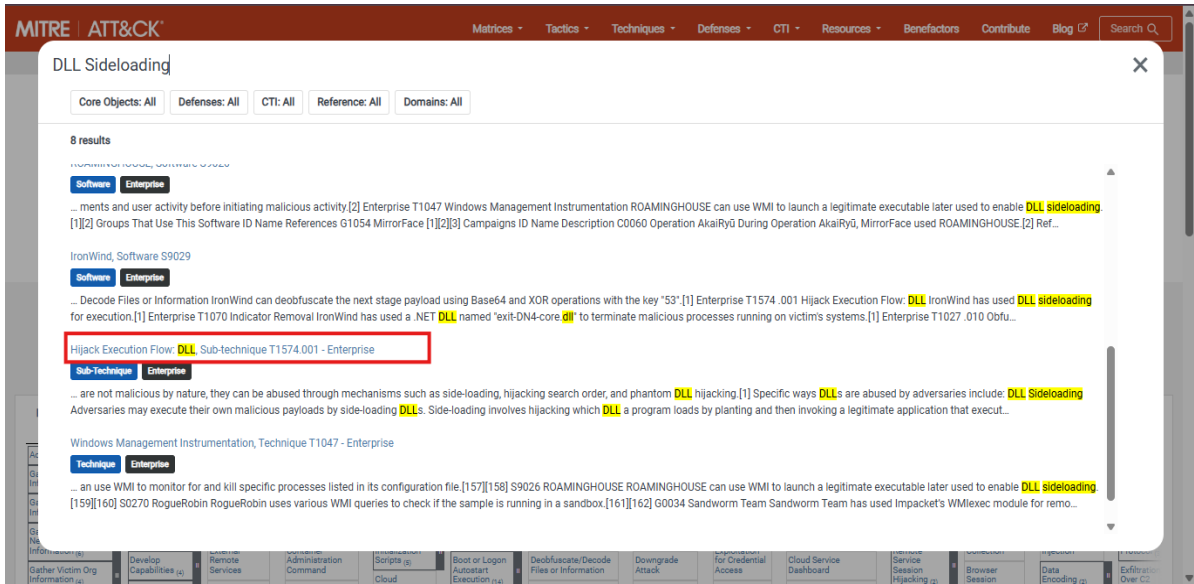


Figure 3 — Supporting reference used to validate PowerShell detection logic and log source mapping.

4. Case Study 2 — DLL Sideloading (T1574.001)

The second case study examined DLL Sideloading, a Hijack Execution Flow technique in which adversaries place a malicious DLL alongside a legitimate application so that the application loads the malicious library instead of (or in addition to) the intended one — allowing code execution under the guise of a trusted process.

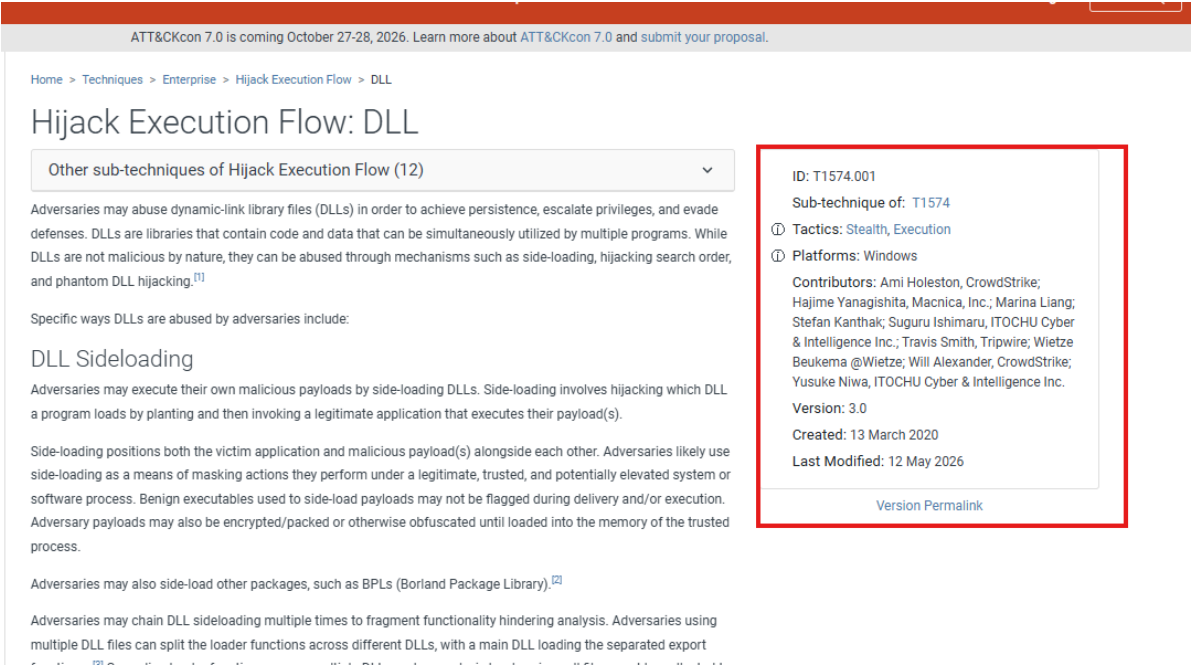


Figure 4 — MITRE ATT&CK technique page for Hijack Execution Flow: DLL Sideloading (T1574.001).

Attack Scenario	DLL Sideloading
Technique Name	Hijack Execution Flow: DLL
Technique ID	T1574.001
ATT&CK Tactic(s)	Defense Evasion, Execution
Platform	Windows

Detection Strategy for DLL Sideloading

MITRE | ATT&CK

Matrices | Tactics | Techniques | Defenses | CTI | Resources | Benefactors | Contribute | Blog | Search

TECHNIQUES

Hijack Execution Flow

DLL

Dylib Hijacking

Executable Installer File Permissions Weakness

Dynamic Linker Hijacking

Path Interception by PATH Environment Variable

Path Interception by Search Order Hijacking

Path Interception by Unquoted Path

Services File Permissions Weakness

Services Registry Permissions Weakness

COR_PROFILER

KernelCallbackTable

AppDomainManager

Input Injection

Inter-Process

M1013 Application Developer Guidance

When possible, include hash values in manifest files to help prevent side-loading of malicious libraries.

M1047 Audit

Use auditing tools capable of detecting DLL search order hijacking opportunities on systems within an enterprise and correct them. Toolkits like the PowerSploit framework contain PowerUp modules that can be used to explore systems for DLL hijacking weaknesses.^[161]

Use the program `assess.exe` that is included with Windows, along with manual inspection, to check manifest files for side-by-side problems in software.^[162]

M1038 Execution Prevention

Identify and block potentially malicious software executed through DLL hijacking by using application control solutions capable of blocking DLLs loaded by legitimate software.^[163]

M1044 Restrict Library Loading

Disallow loading of remote DLLs. This is included by default in Windows Server 2012+ and is available by patch for XP+ and Server 2003+.^[164]

Enable Safe DLL Search Mode to move the user's current folder later in the search order. This is included by default in modern versions of Windows; the associated Windows Registry key is located at `HKEY_LOCAL_MACHINE\CurrentControlSet\Control\Session Manager\SafeDllSearchMode`.^[165]

M1051 Update Software

Update software regularly to include patches that fix DLL side-loading vulnerabilities.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0201	Detection Strategy for Hijack Execution Flow for DLLs	AN0577	DLL hijacking behaviors including unexpected DLL loads from non-standard directories, replacement of DLLs, phantom DLL insertion, redirection file creation, and substitution of legitimate DLLs. Defender correlates file system modifications, registry changes, and module load telemetry to detect abnormal DLL behavior in trusted processes.

References

1. Tom Fakterman, Chen Erlich, & Assaf Dahan. (2024, February 22). Intruders in the

84. Raggi, M. Schwarz, D. (2019, August 1). LookBack Malware Targets the United

Figure 5 — MITRE ATT&CK detection strategy describing how DLL hijacking and DLL sideloading activity can be identified.

Detection Method

Based on the ATT&CK detection guidance, defenders should look for the following behavioral indicators:

- Unexpected DLL loads from non-standard directories
- Replacement of legitimate DLLs
- Phantom DLL insertion
- Redirection file creation
- Substitution of legitimate DLLs
- Correlated file system modifications
- Registry changes associated with search-order or side-loading paths
- Module load telemetry showing abnormal DLLs in trusted processes

Log Source	Why It Is Relevant
Sysmon Event ID 7	Records DLL/Image Load events
Sysmon Event ID 1	Records process creation
Windows Security Logs	Process and security-related events
Microsoft Defender	Detects suspicious DLL activity
EDR/XDR Telemetry	Behavioral detection of DLL hijacking and sideloading
File Integrity Monitoring (FIM)	Detects unexpected DLL file changes
Registry Monitoring Logs	Detects registry modifications used in hijacking

Attack Scenario	ATT&CK Tactic	Technique ID	Detection Method	Relevant Logs
DLL Sideload	Defense Evasion, Execution	T1574.001	Monitor unexpected DLL loads from non-standard directories, DLL replacement, phantom DLL insertion, registry and file system changes, and abnormal module loading in trusted processes	Sysmon Event ID 7, Sysmon Event ID 1, Windows Security Logs, Microsoft Defender, EDR/XDR Telemetry, File Integrity Monitoring, Registry Monitoring Logs

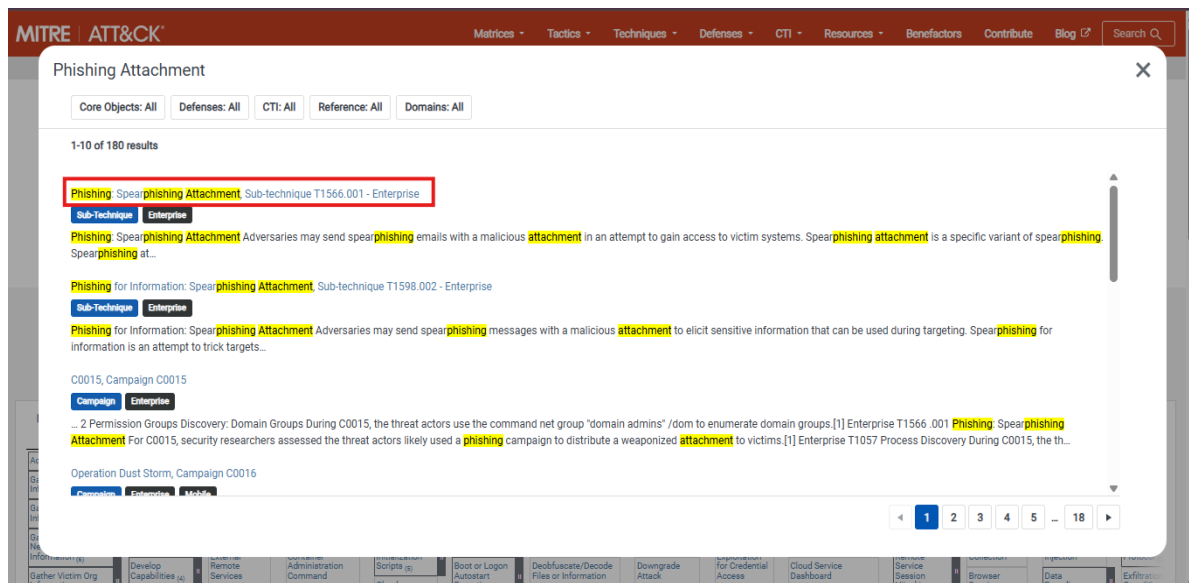


Figure 6 — Reference material used while transitioning from worked case studies to the full 20-technique matrix.

5. Master Mapping — 20 Attack Techniques Across the Attack Lifecycle

Having validated the research-to-detection workflow through the two worked case studies above, the same methodology was applied at scale to 20 attack scenarios spanning the full adversary lifecycle — from Initial Access and Execution through Persistence, Lateral Movement, Credential Access, Discovery, and Impact. Each row below reflects an independently researched ATT&CK technique, its tactic classification, and a practical detection method grounded in a specific, realistic log source.

#	Attack Scenario	ATT&CK Tactic	Technique ID	Detection Method	Relevant Logs
1	PowerShell	Execution	T1059.001	Encoded commands, unusual parent/child processes	PowerShell, Sysmon EID 1, Win 4688, Defender, EDR
2	DLL Sideload	Defense Evasion, Execution	T1574.001	Unexpected DLL loads, registry/file changes	Sysmon EID 7/1, FIM, Defender
3	Spearphishing Attachment	Initial Access	T1566.001	Malicious attachments	Email, Defender, EDR
4	Spearphishing Link	Initial Access	T1566.002	Malicious URLs	Email, Proxy, DNS
5	Command Prompt	Execution	T1059.003	Suspicious cmd.exe activity	Sysmon, Win 4688

#	Attack Scenario	ATT&CK Tactic	Technique ID	Detection Method	Relevant Logs
6	Windows Management Instrumentation	Execution	T1047	Abnormal WMI execution	WMI Logs, Sysmon
7	Scheduled Task	Persistence	T1053.005	Task creation	Win 4698, Sysmon
8	Registry Run Keys	Persistence	T1547.001	Autorun changes	Registry, Sysmon
9	Remote Desktop Protocol	Lateral Movement	T1021.001	Unusual RDP activity	Win 4624 / 4625
10	SMB Admin Shares	Lateral Movement	T1021.002	Admin share access	SMB Logs
11	Valid Accounts	Persistence, Defense Evasion	T1078	Abnormal logins	Win 4624, Active Directory
12	Credential Dumping	Credential Access	T1003	LSASS access	Sysmon, EDR
13	Brute Force	Credential Access	T1110	Repeated login failures	Win 4625
14	Network Service Scanning	Discovery	T1046	Port scanning activity	Firewall, IDS
15	File & Directory Discovery	Discovery	T1083	Enumeration commands	Sysmon
16	System Information Discovery	Discovery	T1082	System query commands	Windows Logs
17	Process Discovery	Discovery	T1057	Process enumeration	Sysmon
18	System Owner/User Discovery	Discovery	T1033	User enumeration	Sysmon
19	Data Encrypted for Impact	Impact	T1486	Mass file encryption	Defender, FIM
20	Inhibit System Recovery	Impact	T1490	Shadow copy deletion	Sysmon, Windows Logs

6. Analysis & Key Observations

6.1 Sysmon as the Common Detection Backbone

Across the 20 mapped techniques, Sysmon event data (process creation, image/DLL loads) appears as a relevant log source far more often than any other single tool — spanning Execution, Persistence, Lateral Movement, Credential Access, and Discovery techniques. This reinforces Sysmon deployment and tuned event filtering as one of the highest-leverage investments a SOC can make for detection coverage.

6.2 Discovery Techniques Cluster Around Native Windows Commands

Techniques 14 through 18 (Network Service Scanning, File & Directory Discovery, System Information Discovery, Process Discovery, and System Owner/User Discovery) all rely on native Windows enumeration commands. Individually, each is low-signal and easily mistaken for legitimate administrative activity; correlating multiple discovery techniques occurring in sequence from the same host is a stronger detection heuristic than alerting on any single one in isolation.

6.3 Authentication Logs Anchor Lateral Movement and Credential Access

RDP abuse (T1021.001), Valid Accounts (T1078), and Brute Force (T1110) all converge on Windows authentication event IDs 4624/4625. This makes centralized, well-baselined logon monitoring — including geographically or

behaviorally anomalous logons — a critical control for catching lateral movement and credential-based attacks before they escalate.

6.4 Impact Techniques Require Pre-Encryption Detection

Data Encrypted for Impact (T1486) and Inhibit System Recovery (T1490) both represent late-stage, high-damage techniques. Because detection at this stage is effectively too late to prevent data loss, these two techniques underscore the importance of catching the earlier-stage techniques in this table — execution, persistence, and defense evasion — well before an adversary reaches the impact phase.

7. Conclusion

This lab reinforced a detection-engineering workflow that goes beyond simply reading ATT&CK technique pages: researching a technique, translating it into a concrete detection method, and mapping that method to specific, realistic log sources. The two worked case studies — PowerShell abuse and DLL Sideloads — demonstrated this process in depth, while the 20-technique master table demonstrated that the same workflow scales consistently across the full attack lifecycle.

The resulting reference table is directly usable as an input to SOC detection rule development, alert triage playbooks, or onboarding documentation — reflecting the kind of structured, source-grounded thinking expected of a SOC Analyst L1 role.